Python Programming Course

A 12-Week Curriculum · From Absolute Beginner to Confident Python Developer

TARGET AUDIENCE	DURATION	PREREQUISITES	PRIMARY GOAL
Absolute beginners — no prior coding experience	12 Weeks (3 Months)	Basic computer literacy + a computer for Python/IDE setup	Build strong Python fundamentals and apply them to real projects, automation and data analysis

MONTH 1
Fundamentals & Core Concepts

W1 Intro to Programming
Python overview & setup, choosing an IDE (VS Code/PyCharm), first program, syntax, variables & data types, input/output.

W2 Operators & Control Flow
Arithmetic, relational & logical operators; if/elif/else; for & while loops; break & continue.

W3 Functions
Defining & calling functions, parameters & arguments, return statement, variable scope, lambda functions.

W4 Core Data Structures
Lists, tuples, dictionaries, sets, and list comprehensions.

MONTH 2
Intermediate Python & Problem-Solving

W5 File Handling
Reading/writing text files, working with CSV files, working with JSON files.

W6 Errors & Exceptions
Understanding exceptions, try/except/else/finally, raising exceptions, debugging techniques.

W7 Object-Oriented Programming
Classes & objects, the __init__ constructor, attributes & methods, inheritance, encapsulation & polymorphism.

W8 Modules, Packages & Standard Library
Importing & creating modules, key standard libraries (datetime, math, random, os, sys), virtual environments.

MONTH 3
Advanced Topics & Practical Application

W9 Data & Essential Libraries
pip & external packages, NumPy arrays, Pandas DataFrames, basic data cleaning & analysis.

W10 APIs & Databases
Consuming APIs with requests, handling JSON responses, SQLite basics, CRUD operations.

W11 Mini Projects (incl. Automation)
To-Do List app · Weather App (API) · Automation Suite — scheduled file/folder organizer & auto email/report sender · Expense Tracker.

W12 Capstone Project & Wrap-up
End-to-end build (data dashboard, web scraper or automation bot), final presentation, course recap & next steps.

ASSESSMENT METHODS

- **Weekly Assignments & Quizzes:** Reinforce topics covered each week
- **Hands-on Labs:** Practical in-class exercises
- **Mini-Projects:** Applied projects at the end of each month
- **Final Capstone Project:** Comprehensive evaluation of all skills learned

SUGGESTED SPECIALIZATION TRACKS (MONTH 3)

- **Data Science:** Deeper NumPy/Pandas, Matplotlib/Seaborn visualization, basic statistics
- **Web Development:** Flask or Django, HTML/CSS basics, deploying a simple web app
- **Automation:** File/OS scripting, task scheduling, email & report automation, browser automation (Selenium)